Hardware, with context.

Eleven documented models and engineering platforms. A technical reference drawn from selected company-supplied source versions.

Use this handbook to compare architectures and request a revision-specific engineering package. It contains selected public parameters; application-specific specifications are available from the team. It is not an operating manual, independent test report, certification or sales quotation.

READ THE EVIDENCE LABELS

Reported - Transcribed from the selected source. No independent performance verification is implied.

Target - A development baseline or design goal, not a measured production specification.

Pending - A value, release or unit has not been established.

Conflict - The supplied source contains incompatible statements.

MODEL INDEX

Micro drives
M01, M01 Neo, M02

Large joints
WZ53, WZ79, WZ119

Dexterous hands
Neo hand, Dexterous hand 21

Robot platforms
Quadruped platform, Spherical R700, Dual-arm platform

Source reference: MP-PRODUCTS-2026.10-r1
Public edition: 14 October 2026
Documentation: flat-hill.github.io/company/docs/
Product center: flat-hill.github.io/company/products/

M01

Micro drives | M01 is a miniature rotary actuator architecture for finger mechanisms and compact robotic motion.

Parameter	Value & context	Basis / source
Rated torque	0.5 N·m	Reported S1 · slide 9
Stall torque	1.5 N·m Stall torque is not a continuous operating rating.	Reported S1 · slide 9
Rated speed	113 rpm	Reported S1 · slide 9
Envelope	29 × 23 × 17.5 mm	Reported S1 · slide 9
Supply voltage	12 V	Reported S1 · slide 9
Maximum bus current	2 A	Reported S1 · slide 9
Reduction ratio	220:1	Reported S1 · slide 9
Transmission	Planetary gearbox	Reported S1 · slide 9
Motor	PMSM brushless	Reported S1 · slide 9
Communication	RS-485 / CAN Protocol, pinout and firmware version require confirmation.	Reported S1 · slide 9
Feedback	Dual magnetic encoders	Reported S1 · slide 9
Mass	Not published	Pending S1 · slide 9

VERSION NOTES

Values are transcribed from the selected source version, not independently verified acceptance specifications. Confirm the exact hardware revision, operating conditions and availability with the team.

Stall and rated torque must be evaluated separately. Do not assume that the maximum torque and rated speed occur simultaneously.

Use the slide 9 envelope for this published snapshot. The later iteration roadmap uses a different first-generation envelope; confirm the drawing revision before mechanical integration.

View the technical page and parameter CSV online

M01 Neo

Micro drives | M01 Neo is a miniature rotary actuator architecture for finger mechanisms and compact robotic motion.

Parameter	Value & context	Basis / source
Rated torque	0.3 N·m	Reported S1 · slide 18
Stall torque	0.9 N·m Stall torque is not a continuous operating rating.	Reported S1 · slide 18
Rated speed	168 rpm	Reported S1 · slide 18
Envelope	31 × 18 × 16 mm	Reported S1 · slide 18
Supply voltage	12 V	Reported S1 · slide 18
Maximum bus current	2 A	Reported S1 · slide 18
Reduction ratio	107:1	Reported S1 · slide 18
Transmission	Planetary + fixed-axis gears	Reported S1 · slide 18
Motor	PMSM brushless	Reported S1 · slide 18
Communication	RS-485 / CAN Protocol, pinout and firmware version require confirmation.	Reported S1 · slide 18
Feedback	Dual magnetic encoders	Reported S1 · slide 18
Mass	Not published	Pending S1 · slide 18

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View the technical page and parameter CSV online

M02

Micro drives | M02 is a miniature rotary actuator architecture for finger mechanisms and compact robotic motion.

Parameter	Value & context	Basis / source
Rated torque	1.2 N·m	Reported S1 · slide 10
Stall torque	3.6 N·m Stall torque is not a continuous operating rating.	Reported S1 · slide 10
Rated speed	89 rpm	Reported S1 · slide 10
Envelope	40 × 22 × 16 mm	Reported S1 · slide 10
Supply voltage	12 V	Reported S1 · slide 10
Maximum bus current	3 A	Reported S1 · slide 10
Reduction ratio	84:1	Reported S1 · slide 10
Transmission	Planetary gearbox	Reported S1 · slide 10
Motor	PMSM brushless	Reported S1 · slide 10
Communication	RS-485 / CAN Protocol, pinout and firmware version require confirmation.	Reported S1 · slide 10
Feedback	Dual magnetic encoders	Reported S1 · slide 10
Mass	Not published	Pending S1 · slide 10

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Stall and rated torque must be evaluated separately. Do not assume that the maximum torque and rated speed occur simultaneously.

This M02 entry describes the first-generation D19 actuator in the August 2026 presentation. It does not establish the release status of a later roadmap product using the same name.

View the technical page and parameter CSV online

WZ53

Large joints | A 53 mm joint-module design baseline with a 7 N·m rated-output target, integrating transmission, position sensing and drive electronics.

Parameter	Value & context	Basis / source
Rated torque target	7 N·m	Target S2 · pages 1–3
Outer diameter	53 mm	Target S2 · pages 1–3
Mass target	300 g	Target S2 · pages 1–3
Supply voltage	48 V	Target S2 · pages 1–3
Reduction ratio	12–20:1 Final ratio requires calculation and prototype testing.	Target S2 · pages 1–3
Transmission	Single-stage NW	Target S2 · pages 1–3
Motor baseline	5013 inner-rotor frameless Purchased motor in the prototype baseline.	Target S2 · pages 1–3
Shaft architecture	Non-hollow	Target S2 · pages 1–3
Communication	CAN	Target S2 · pages 1–3
Feedback	Input + output position sensing	Target S2 · pages 1–3
Rated speed	Pending unit confirmation Source lists 80 without a unit; no rpm value is published here.	Pending S2 · page 3
Production qualification	Not established by this source	Pending S2 · pages 1–2

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A self-developed module architecture can incorporate purchased motors and electronic components.

View the technical page and parameter CSV online

WZ79

Large joints | A 79 mm joint-module design baseline with a 15 N·m rated-output target, integrating transmission, position sensing and drive electronics.

Parameter	Value & context	Basis / source
Rated torque target	15 N·m	Target S2 · pages 1–3
Outer diameter	79 mm	Target S2 · pages 1–3
Mass target	500 g	Target S2 · pages 1–3
Supply voltage	48 V	Target S2 · pages 1–3
Reduction ratio	12–20:1 Final ratio requires calculation and prototype testing.	Target S2 · pages 1–3
Transmission	Single-stage NW	Target S2 · pages 1–3
Motor baseline	7613 inner-rotor frameless Purchased motor in the prototype baseline.	Target S2 · pages 1–3
Shaft architecture	Non-hollow	Target S2 · pages 1–3
Communication	CAN	Target S2 · pages 1–3
Feedback	Input + output position sensing	Target S2 · pages 1–3
Rated speed	Pending unit confirmation Source lists 80 without a unit; no rpm value is published here.	Pending S2 · page 3
Production qualification	Not established by this source	Pending S2 · pages 1–2

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A self-developed module architecture can incorporate purchased motors and electronic components.

View the technical page and parameter CSV online

WZ119

Large joints | A 119 mm joint-module design baseline with a 60 N·m rated-output target, integrating transmission, position sensing and drive electronics.

Parameter	Value & context	Basis / source
Rated torque target	60 N·m	Target S2 · pages 1–3
Outer diameter	119 mm	Target S2 · pages 1–3
Mass target	1,000 g	Target S2 · pages 1–3
Supply voltage	48 V	Target S2 · pages 1–3
Reduction ratio	12–20:1 Final ratio requires calculation and prototype testing.	Target S2 · pages 1–3
Transmission	Single-stage NW	Target S2 · pages 1–3
Motor baseline	11513 inner-rotor frameless Purchased motor in the prototype baseline.	Target S2 · pages 1–3
Shaft architecture	Large hollow shaft	Target S2 · pages 1–3
Communication	CAN	Target S2 · pages 1–3
Feedback	Input + output position sensing	Target S2 · pages 1–3
Rated speed	Pending unit confirmation Source lists 80 without a unit; no rpm value is published here.	Pending S2 · page 3
Production qualification	Not established by this source	Pending S2 · pages 1–2

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A self-developed module architecture can incorporate purchased motors and electronic components.

View the technical page and parameter CSV online

Neo hand

Dexterous hands | A five-finger, 20-DOF fully actuated hand architecture for manipulation research and system integration.

Parameter	Value & context	Basis / source
Active degrees of freedom	20	Reported S1 · slide 23
Finger count	5	Reported S1 · slide 23
Body mass	900 g	Reported S1 · slide 23
Rated pinch force	20 N Source terminology; test geometry and conditions need confirmation.	Reported S1 · slide 23
Rated enveloping grasp force	60 N	Reported S1 · slide 23
Supply voltage	12 V	Reported S1 · slide 23
Maximum current	15 A	Reported S1 · slide 23
Communication	RS-485	Reported S1 · slide 23
Control frequency	320 Hz	Reported S1 · slide 23
CAD / firmware release	Request the matching revision	Pending S1 · slide 23

VERSION NOTES

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Pinch force and enveloping grasp force are different metrics; neither is a payload rating.

The selected August presentation states 900 g, 20 N pinch, 60 N grasp and 320 Hz. Other promotional materials contain different values; they are not merged into this version.

View the technical page and parameter CSV online

Dexterous hand 21

Dexterous hands | A five-finger, 21-DOF fully actuated hand architecture for manipulation research and system integration.

Parameter	Value & context	Basis / source
Active degrees of freedom	21	Reported S1 · slide 15
Finger count	5	Reported S1 · slide 15
Body mass	1,650 g	Reported S1 · slide 15
Rated pinch force	40 N Source terminology; test geometry and conditions need confirmation.	Reported S1 · slide 15
Rated enveloping grasp force	120 N	Reported S1 · slide 15
Supply voltage	12 V	Reported S1 · slide 15
Maximum current	20 A	Reported S1 · slide 15
Communication	RS-485	Reported S1 · slide 15
Control frequency	320 Hz	Reported S1 · slide 15
CAD / firmware release	Request the matching revision	Pending S1 · slide 15

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Pinch force and enveloping grasp force are different metrics; neither is a payload rating.

View the technical page and parameter CSV online

Quadruped platform

Robot platforms | An engineering platform for locomotion, perception and mobile robotics development.

Parameter	Value & context	Basis / source
Total degrees of freedom	16 12 leg joints + 4 configuration-dependent axes; confirm wheel/gimbal allocation.	Reported S3 · page 3
Standing envelope	700 × 467 × 645 mm Without exterior cover.	Reported S3 · page 3
Folded envelope	700 × 600 × 238 mm Without exterior cover.	Reported S3 · page 3
Mass with battery	~29 kg	Reported S3 · page 3
Mass without battery	~24 kg	Reported S3 · page 3
Supply voltage	48 V	Reported S3 · page 3
Maximum joint torque	130 N·m Not a continuous joint rating.	Reported S3 · page 3
Motion speed	0–3 m/s	Reported S3 · page 3
Maximum slope	45° Terrain, loading and test protocol require confirmation.	Reported S3 · page 3
External ports	1× GbE / 2× USB 3.0 / 2× Type-C / 1× HDMI	Reported S3 · page 3
Compute expansion	Jetson Orin Nano / NX option Optional expansion, not a guaranteed installed module.	Reported S3 · page 10

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The extra four degrees of freedom are configuration-dependent in the manual. The platform illustration does not freeze their implementation.

Runtime, payload, sensing and SDK delivery must be agreed for the selected configuration.

Selected public parameters. Application-specific specifications are available from the team.

View the technical page and parameter CSV online

Spherical R700

Robot platforms | A 700 mm spherical robot design with internal pendulum drive and flywheel stabilization.

Parameter	Value & context	Basis / source
Body diameter	700 mm	Target S4 · page 2
System mass	90 kg	Target S4 · page 2
Payload	15 kg	Target S4 · page 2
Supply voltage	48 V	Target S4 · page 2
Maximum speed	25 km/h	Target S4 · page 3
Cruise speed	10 km/h	Target S4 · page 3
Maximum slope	17° Hard-surface target; wet terrain requires derating.	Target S4 · page 3
Obstacle height	Unresolved: 100 / 200 mm Motion table says 100 mm; summary says 200 mm. Do not use either as an accepted limit.	Conflict S4 · pages 3–4
Interfaces	Power / Ethernet / CAN / serial	Target S4 · page 4

VERSION NOTES

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The brief mixes target and descriptive language. Performance and environmental figures are treated as design targets until test evidence is available.

Obstacle-height values conflict within the source. An agreed acceptance test and corrected specification are required.

Selected public parameters. Application-specific specifications are available from the team.

View the technical page and parameter CSV online

Dual-arm platform

Robot platforms | A modular dual-arm research architecture for coordinated manipulation and embodied-AI workflows.

Parameter	Value & context	Basis / source
Degrees of freedom	14 total / 7 per arm	Reported S5 · page 1
Mass per arm	5.5 kg	Reported S5 · page 1
Reach per arm	633 mm	Reported S5 · page 1
Joint allocation per arm	3 shoulder / 1 elbow / 3 wrist	Reported S5 · page 1
Joint interface	CAN	Reported S5 · page 1
Structure	Aluminum + printed composite covers	Reported S5 · page 1
Control architecture	QDD / compliant-control concept	Reported S5 · page 1
Software references	ROS 2 / Python / MuJoCo / Isaac Sim Listed by the source; release artifacts and compatibility are to be confirmed.	Reported S5 · page 1
Rated payload	Not published	Pending S5 · page 1
Repeatability	Not published	Pending S5 · page 1
Released SDK / CAD	Request access and license terms	Pending S5 · page 1

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The source mentions open resources, but no current public SDK, CAD release or license is established by the reviewed materials.

Do not infer a human-collaboration safety certification from a compliant-control architecture.

View the technical page and parameter CSV online

Integration & source register.

Agree the model, hardware revision, interfaces and acceptance conditions before committing to a build.

Mechanical - Envelope, mounting pattern, load cases, cable exits and tolerances.

Electrical - Voltage range, continuous and peak current, connectors, grounding and bus termination.

Software - Protocol, firmware, SDK, simulation model and license terms.

Validation - Duty cycle, test environment, measurements, thermal limits and release evidence.

SOURCE REGISTER

S1 / Micro Dynamics Actuator & Dexterous Hand
August 2026 presentation
Supplied product presentation

S2 / Joint Module Prototype Development Plan
Supplied copy; revision date not stated
Prototype design targets

S3 / Quadruped Inspection Platform Manual
Supplied copy; revision date not stated
Supplied product manual

S4 / Spherical Mobile Robot WZ-R700
Supplied copy; revision date not stated
Engineering product brief

S5 / Embodied Intelligence Dual-arm Platform
Supplied one-page brief; revision date not stated
Supplied product brief

The source register identifies internal documents used for the technical extract. Full documents are not republished. Values from different generations are kept separate; the spherical platform obstacle-height conflict is left unresolved. Released CAD, SDK and test packages require team confirmation.